Year 11 Set 2 · Higher · grade 5–6 · Lesson 8

Indices & standard form

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Today I will learn to...

Goals

1. Use negative and fractional indices

2. Calculate with numbers in standard form

Do now: retrieval practice

4 mins

1 2^5

32

2 $\sqrt{(9 \times 4)}$

6

3 Expand $(x + 3)(x - 3)$

$x^2 - 9$

4 $F \propto 1/d$, $F = 8$ when $d = 2$. F when $d = 4$?

4

Objective 1

Use negative and fractional indices

Key idea

Learn

!

$a^{-n} = 1/a^n$. $a^{1/n}$ is the nth root. $a^{m/n}$: take the root first, then the power.

Objective 1: Use negative and fractional indices

I do: example 1

I do

3^{-2}

1. Negative power means 1 over: $3^{-2} = 1/3^2$
2. $3^2 = 9$
3. $= 1/9$
4. Check: $3^2 \times 3^{-2} = 3^0 = 1$ ✓

I do: example 2

I do

$27^{2/3}$

1. Denominator 3 → cube root
2. $\sqrt[3]{27} = 3$
3. Numerator 2 → square it: 3^2
4. $= 9$

I do: example 3

I do

$(16/81)^{-1/2}$

1. Negative power → flip the fraction: $(81/16)^{1/2}$
2. Power ½ → square root
3. $\sqrt{81} = 9$ and $\sqrt{16} = 4$
4. $= 9/4$

Mini whiteboards

We do

1. $64^{1/3}$

Answer: 4

2. $8^{-2/3}$

Answer: 1/4

3. $25^{-1/2}$

Answer: 1/5

4. $(1/8)^{-1/3}$

Answer: 2

You do: choose your challenge

You do

Bronze

Work out 2^{-3}

Silver

Work out $16^{3/4}$

Gold

Simplify $x^5 \times x^{-2} \div x^4$

Platinum

Find n if $2^n = 8^3 \times 4$

You do: answers

Check

Bronze

1/8

Silver

8

Gold

$x^{-1} = 1/x$

Platinum

$2^9 \times 2^2 = 2^{11}$, n = 11

If you finish early

Extra

Bronze

Work out $9^{1/2}$

3

Silver

Work out $27^{-1/3}$

1/3

Gold

Work out $(16/25)^{3/2}$

64/125

Platinum

Simplify $(27x^{6})^{2/3}$

$9x^{4}$

Exam question 1

2 marks

Work out $125^{-2/3}$

Space for working

Mark scheme: 1/25

Exam question 2

2 marks

Simplify $(3a^2b)^3$

Space for working

Mark scheme: $27a^6b^3$

Objective 1 achieved

Use negative and fractional indices



Objective 2

Calculate with numbers in standard form

✓ Achieved: Use negative and fractional indices

Key idea

Learn

!

Standard form is $A \times 10^n$ with $1 \leq A < 10$. To multiply, multiply the numbers and add the powers. To divide, divide the numbers and subtract the powers. Then make sure A is between 1 and 10.

Objective 2: Calculate with numbers in standard form

I do: example 1

I do

$(3 \times 10^4) \times (2 \times 10^5)$

1. $3 \times 2 = 6$
2. $10^{4+5} = 10^9$
3. 6×10^9

I do: example 2

I do

$(8 \times 10^7) \div (4 \times 10^2)$

1. $8 \div 4 = 2$
2. $10^{7-2} = 10^5$
3. 2×10^5

I do: example 3

I do

$(5 \times 10^{3}) \times (6 \times 10^{-8})$

1. $5 \times 6 = 30$
2. $10^{3-8} = 10^{-5}$
3. $30 \times 10^{-5} = 3 \times 10^{-4}$

Mini whiteboards

We do

1. Write 0.00042 in standard form

Answer: 4.2×10^{-4}

2. $(2 \times 10^{3}) \times (4 \times 10^{6})$

Answer: 8×10^{9}

3. Write 5 600 000 in standard form

Answer: 5.6×10^{6}

4. $(6 \times 10^{8}) \div (2 \times 10^{3})$

Answer: 3×10^{5}

You do: choose your challenge

You do

Bronze

Write 34 000 in standard form

Silver

$(4 \times 10^3) \times (3 \times 10^5)$

Gold

$(9 \times 10^4) \div (3 \times 10^{-3})$

Platinum

A virus is 1.2×10^{-7} m long. How many would fit end to end along 3 mm?

You do: answers

Check

Bronze

3.4×10^4

Silver

$12 \times 10^8 = 1.2 \times 10^9$

Gold

3×10^7

Platinum

$3 \times 10^{-3} \div 1.2 \times 10^{-7} = 2.5 \times 10^4$

If you finish early

Extra

Bronze

Write 7.3×10^{-3} as an ordinary number.

0.0073

Silver

Work out $(5 \times 10^{4}) \times (3 \times 10^{2})$ in standard form.

1.5×10^{7}

Gold

Work out $(8.4 \times 10^{5}) \div (2 \times 10^{-2})$ in standard form.

4.2×10^{7}

Platinum

The Sun is 1.5×10^{8} km from Earth. Light travels 3×10^{5} km per second. How many seconds does sunlight take to reach Earth?

500 seconds

Exam question 1

2 marks

Work out $(6 \times 10^5) \div (3 \times 10^{-2})$. Give your answer in standard form.

Space for working

Mark scheme: 2×10^7

Exam question 2

2 marks

The Sun is 1.5×10^8 km from Earth. Light travels 3×10^5 km per second. How many seconds does sunlight take to reach Earth?

Space for working

Mark scheme: $1.5 \times 10^8 \div 3 \times 10^5 = 500$ seconds

Objective 2 achieved

Calculate with numbers in standard form



Real life

Apply

A grain of sand has a mass of 6.7×10^{-4} g. What is the mass of 2 million grains, in kg?

$6.7 \times 10^{-4} \times 2 \times 10^{6} = 1\,340$ g = 1.34 kg

Extension: stretch yourself

Stretch

1. Find n if $9^n = 3^8$

$3^{2n} = 3^8$, n = 4

2. Rationalise $6/\sqrt{3}$ (stretch to grade 7)

$6\sqrt{3}/3 = 2\sqrt{3}$

Exit ticket: exam questions

Exit

1. Work out $32^{2/5}$ **(2 marks)**

2. Write 0.00075 in standard form **(1 mark)**

3. Work out $(2.4 \times 10^6) \div (8 \times 10^2)$. Give your answer in standard form. **(2 marks)**

Exit ticket: mark scheme

Mark it

Q1 4 [2]

Q2 7.5×10^{-4} [1]

Q3 $0.3 \times 10^{4} = 3 \times 10^{3}$ [2]

Total: ____ / 5

Today I achieved

- ✔ **Use negative and fractional indices**
- ✔ **Calculate with numbers in standard form**

Rate yourself: 1 = still unsure 2 = getting there 3 = confident

Homework: exam questions

Due next lesson

1. Work out $49^{-1/2}$ [2]

2. Simplify $(2x^3)^2 \times 3x$ [2]

3. Write 6 080 000 in standard form [1]

4. Work out $(7 \times 10^5) \times (6 \times 10^{-2})$. Give your answer in standard form. [2]

5. A dust particle has mass 7.5×10^{-10} kg. How many particles have a total mass of 3×10^{-3} kg? [2]

Extra practice: Maths Genie → Edexcel question bank → Indices · Standard Form